第5次作业 3.39 3.41 3.45 3.47 3.49
3.52 3.57 3.61 3.64

3.39 解 $\begin{cases} mv_0 l = J\omega + mv_1 l \\ J = \frac{1}{3}Ml^2 \\ \frac{1}{2}mv_0^2 = \frac{1}{2}J\omega^2 + \frac{1}{2}mv_1^2 \\ \frac{1}{2}mgl(1-\cos\theta) = \frac{1}{2}J\omega^2 \\ L = J\omega \end{cases} \Rightarrow \begin{cases} \omega = \sqrt{\frac{3g(1-\cos\theta)}{l}} \\ v_0 = \frac{Ml+3ml}{6m}\sqrt{3gl(1-\cos\theta)} \\ L = \frac{1}{3}Ml\sqrt{3gl(1-\cos\theta)} \end{cases}$

3.41 解 $\begin{cases} mv\frac{l}{2} = J\omega \\ J = \frac{1}{12}Ml^2 \end{cases} \Rightarrow \omega = \frac{6mv}{Ml}$

3.45 解 $\begin{cases} F - f = Ma_c \\ fR - F_n\frac{l}{2} = J\beta \\ \beta R = a_c \\ \mu mg = f \end{cases} \Rightarrow \begin{cases} \beta = \frac{R-r}{J+MR^2}F \\ a_c = \frac{R-r}{J+MR^2}FR \\ f = \frac{J+MRr}{J+MR^2}F \\ \mu = \frac{J+MRr}{J+MR^2}\frac{F}{Mg} \end{cases}$

$\Rightarrow \begin{cases} \beta = 10\,rad/s^2 \\ a = 2\,m/s^2 \\ f = 17N \\ \mu = 0.425 \end{cases}$

3.47 $\begin{cases} 4mg - T = ma_c \\ TR = J\beta \\ J = \frac{1}{2}mR^2 \\ \beta R = a_c' \\ \vec{a_c} = \vec{a_c'} + \vec{3g} \end{cases} \Rightarrow \begin{cases} \beta = \frac{8g}{3R} \\ a_c = \frac{1}{3}g \\ T = \frac{4}{3}mg \end{cases}$

3.49 $\begin{cases} T_2r_2 - T_1r_1 = J\beta \\ T_1 - Mg - T_2 = Ma_c \\ a_c = \beta r_1 \\ a_m + a_c = \beta r_2 \\ mg - T_2 = ma_m \end{cases} \Rightarrow \begin{cases} \beta = 6.21\,rad/s \\ a_c = 0.23\,m/s^2 \\ a_m = 0.373\,m/s^2 \\ T_1 = 139.75N \\ T_2 = 57.76N \end{cases}$

3.52 $\begin{cases} mg(h-2R) = \frac{1}{2}mv^2 + \frac{1}{2}J\omega^2 \\ v = \omega r \\ J = \frac{2}{5}mr^2 \\ mg = m\frac{v^2}{R} \end{cases} \Rightarrow h = 2.7R$

即 $h \geq 2.7R$

Scanned with CamScanner

3.57 $T_1 - mg - ma_0 = ma$

$$\begin{cases} T_2R - T_1R = J\beta_1 \\ Mg + Ma_0 - T_2 = M(a_C' + a) \\ T_2R = J\beta_2 \\ a_C' = \beta_2 R \\ a = \beta_1 R \\ J = \frac{1}{2}MR^2 \end{cases}$$

$$\Rightarrow \begin{cases} a_C' = \dfrac{(2M+3m)(g+a_0)}{5M+6m} \\ a = \dfrac{(2M-6m)(g+a_0)}{5M+6m} \end{cases}$$

$$\Rightarrow \begin{cases} a_2 = 2m/s^2 \\ a_C = 8.67m/s^2 \\ T_2 = 20N \\ T_1 = 14N \end{cases}$$

3.61 $L = J\omega$

$J = mR_G^2$

$\Rightarrow L = m\omega R_G^2$

$dL = mg\frac{l}{2}dt$

$dL = L\,d\theta$

$\Rightarrow \Omega = \dfrac{gl}{2\omega R_G^2}$

旋进方向为俯视逆时针

3.64 $P_0 + \frac{1}{2}\rho V_0^2 = P_1$

$P_0 + \rho g h = P_1$

$\Rightarrow h = 0.015m$

Scanned with CamScanner